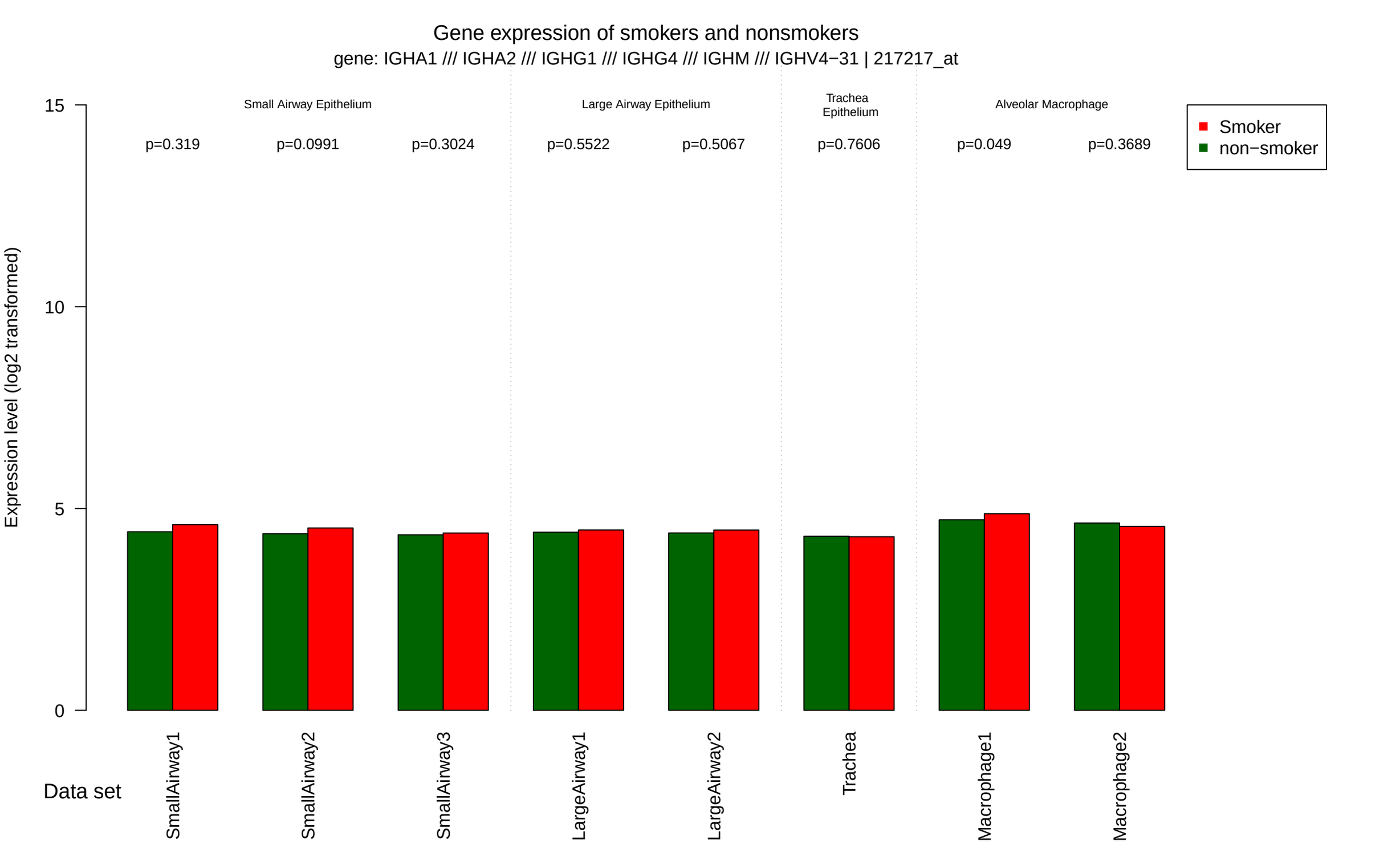
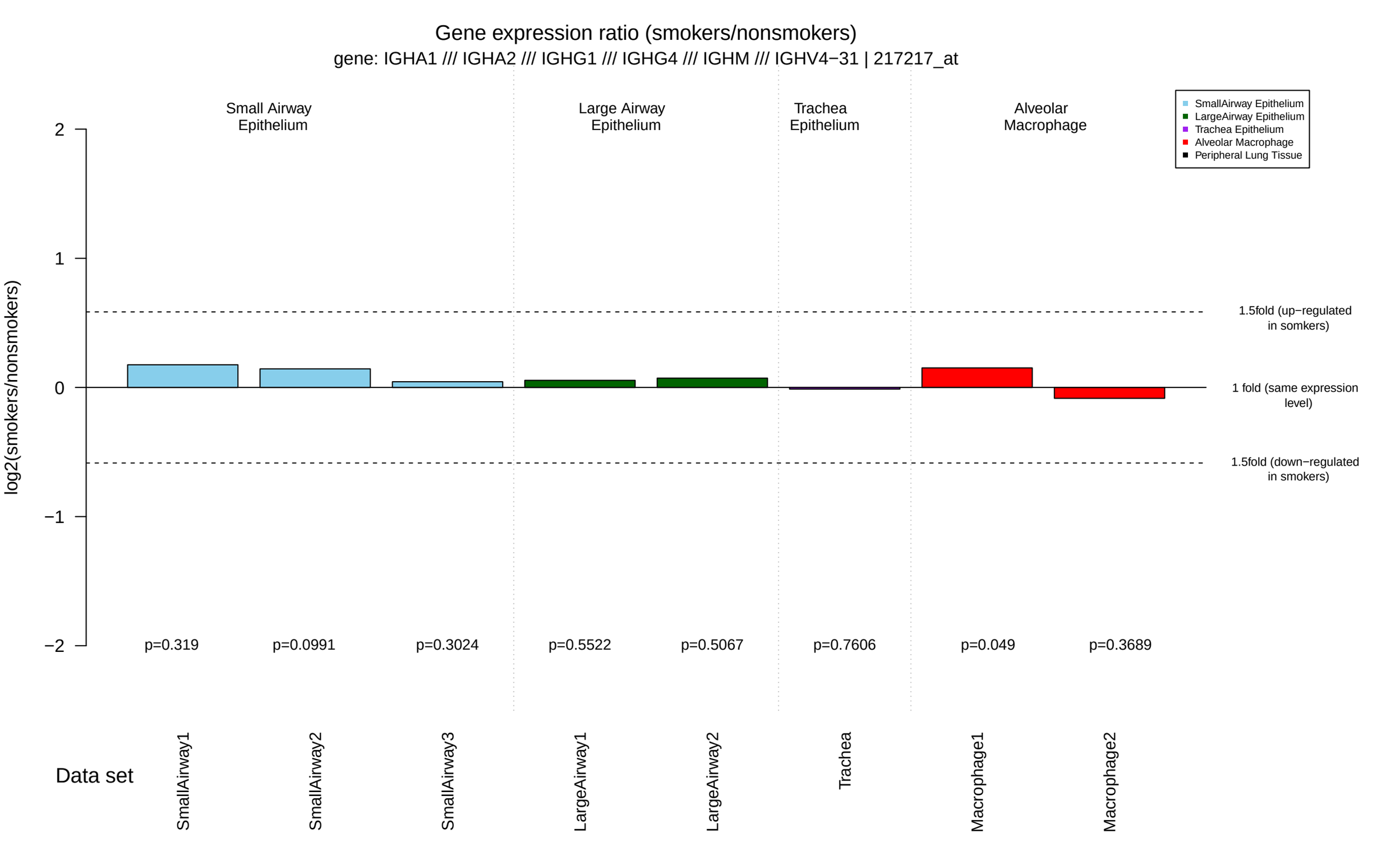
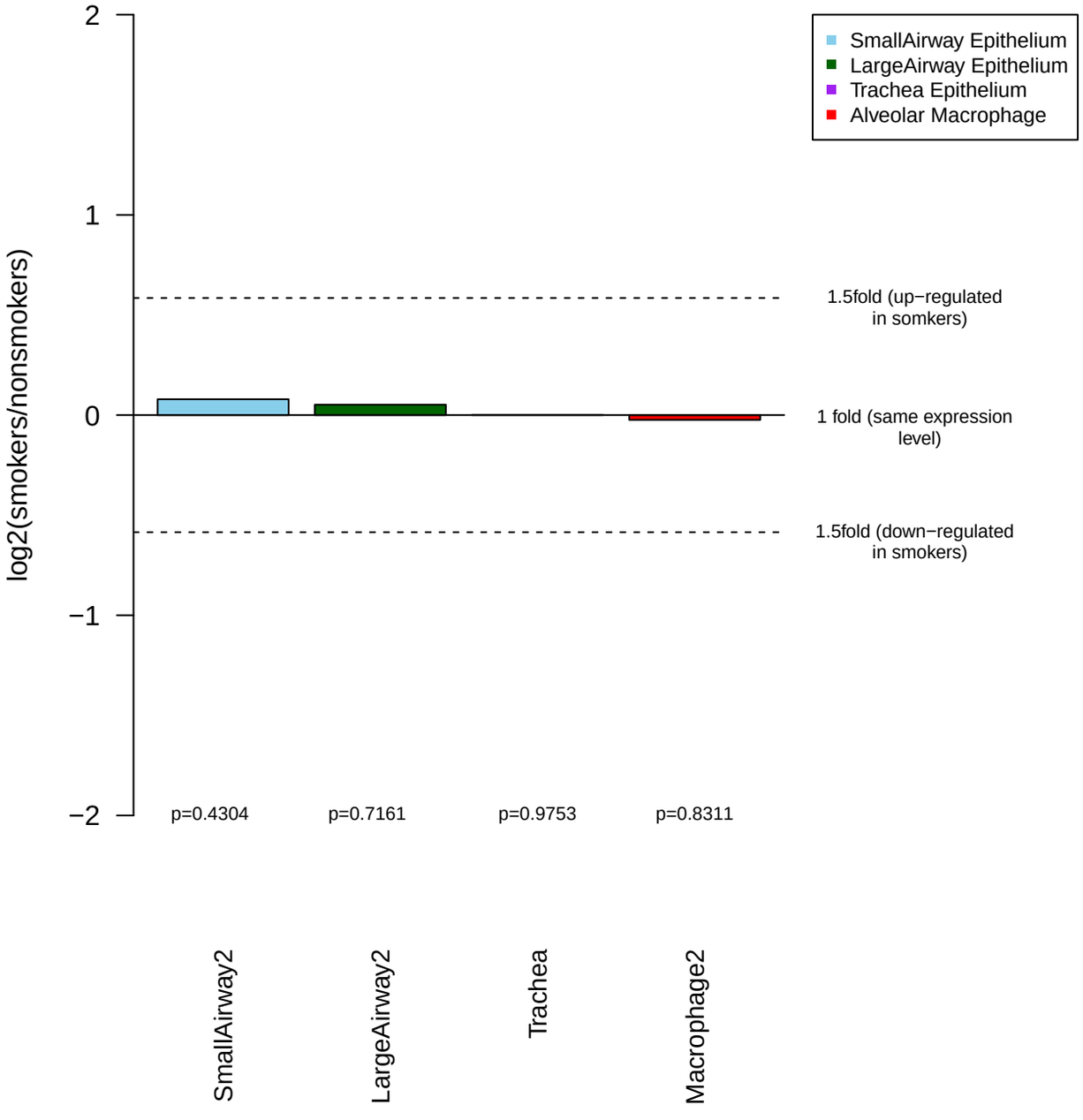

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.319
p=0.0991
p=0.3024
p=0.5522
p=0.5067
p=0.7606
p=0.049
p=0.3689
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.319
p=0.0991
p=0.3024
p=0.5522
p=0.5067
p=0.7606
p=0.049
p=0.3689
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

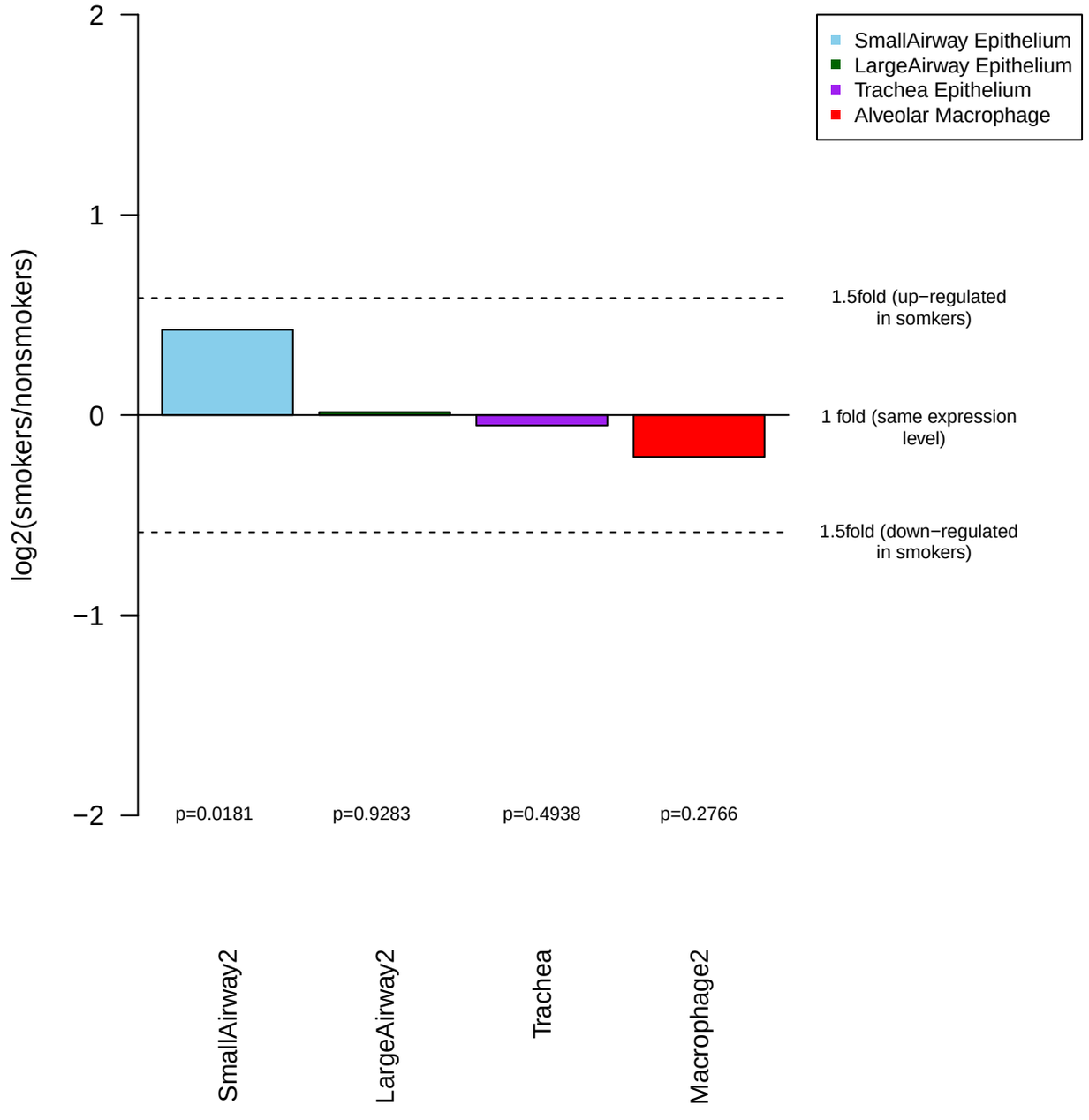
Gene expression ratio in Males

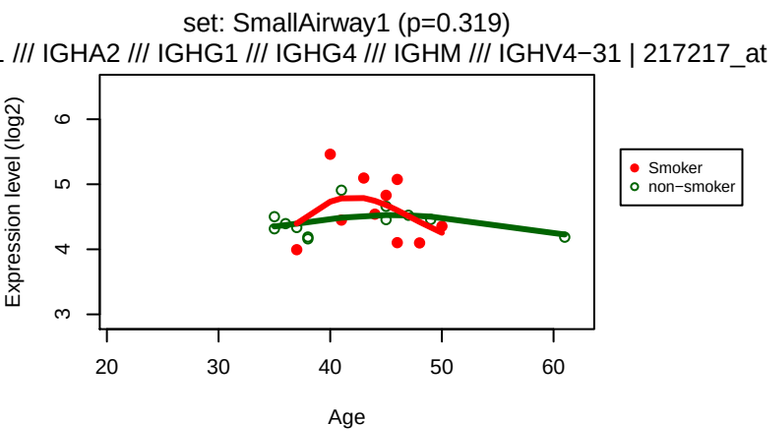
ene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at

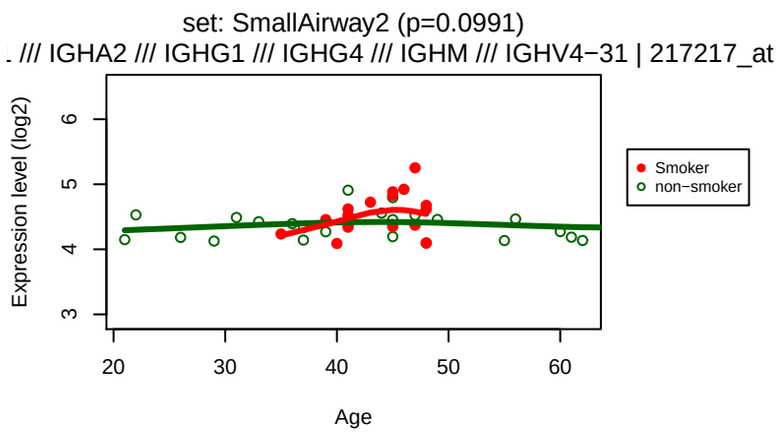


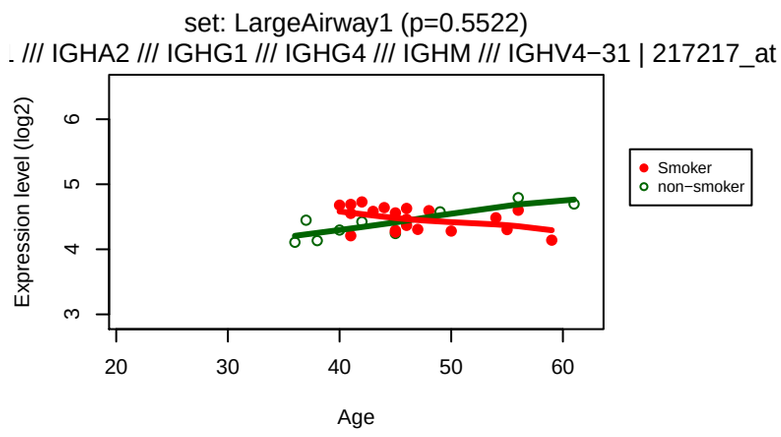
Gene expression ratio in Females

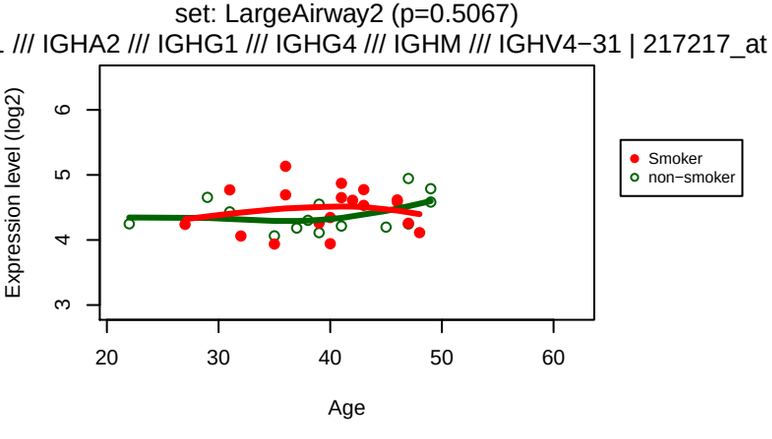
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at

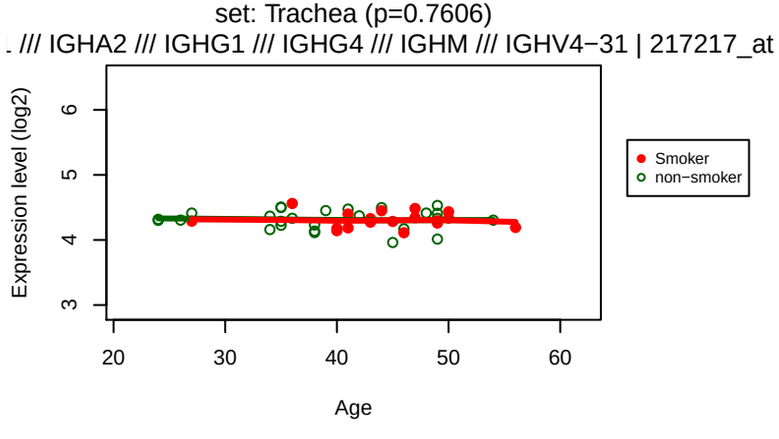


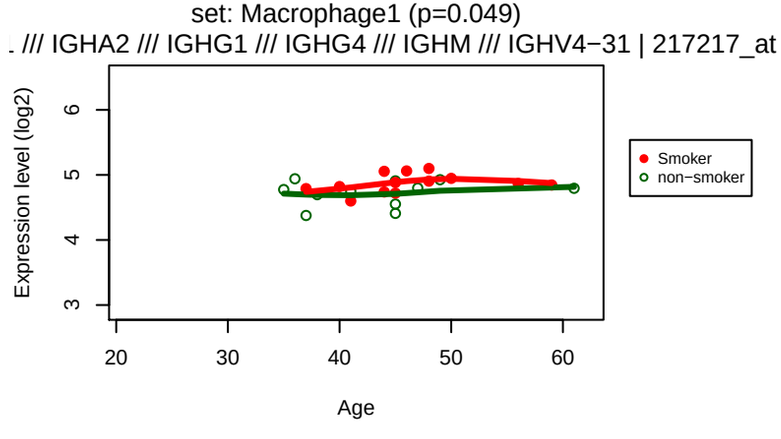

set: SmallAirway1 (p=0.319)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker

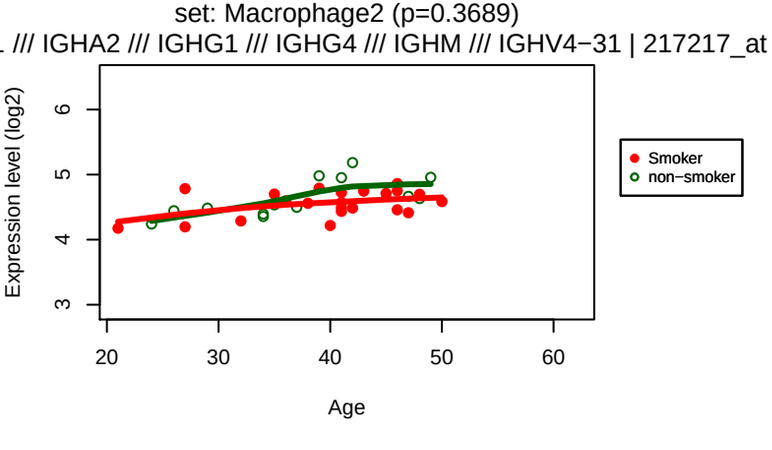

set: SmallAirway2 (p=0.0991)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker

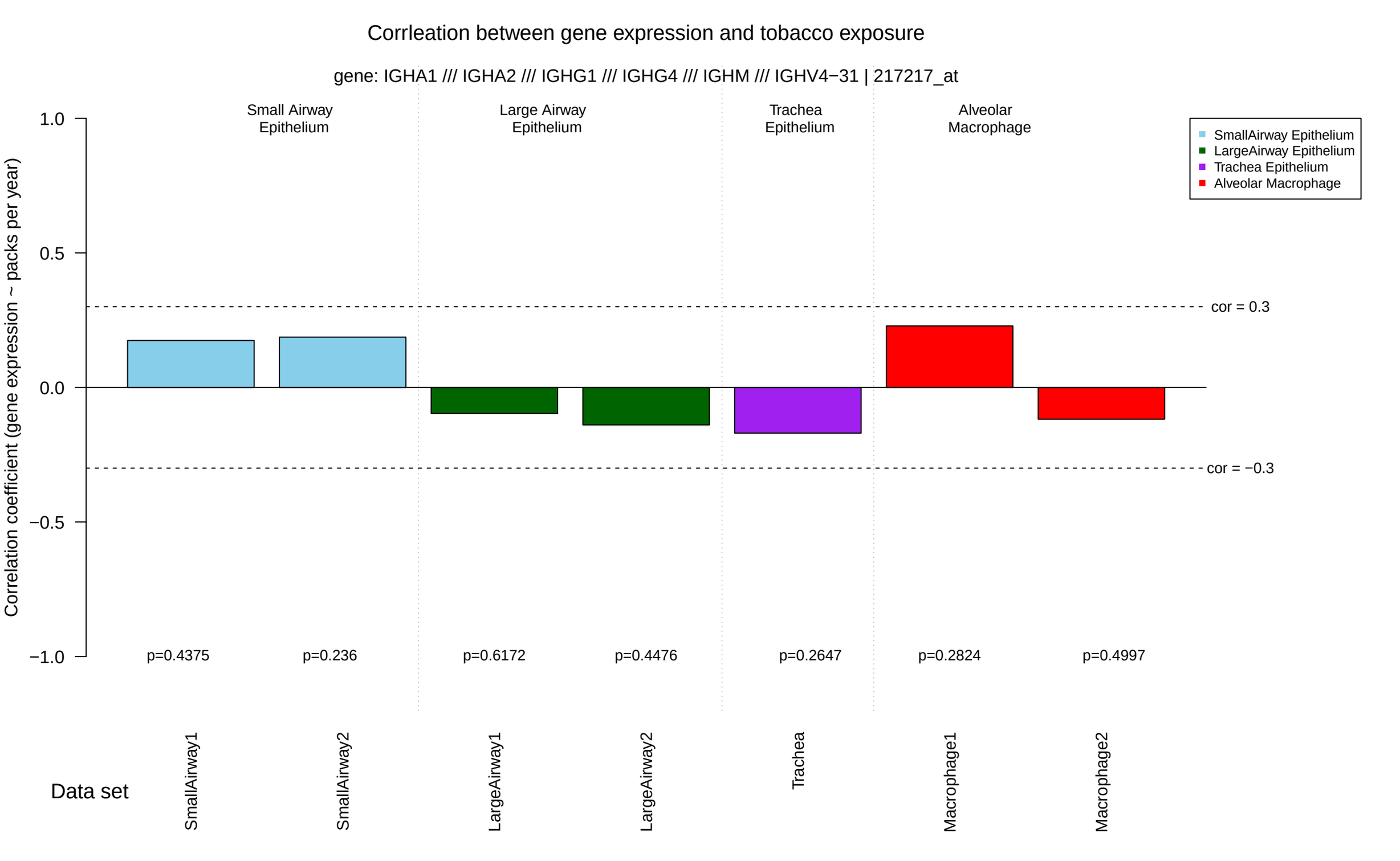

set: LargeAirway1 (p=0.5522)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5067)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7606)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.049)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3689)
/// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 217217_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4375
p=0.236
p=0.6172
p=0.4476
p=0.2647
p=0.2824
p=0.4997
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2